Prosthesis Body Arm

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ABSTRACT

The loss of one hand can significantly affect the level of autonomy and the capability of perform daily living, working and social activities. The current prosthetic solutions contribute in a poor way to overcome these problems due to limitations in the interfaces adopted for controlling the prosthesis and to the lack of force or tactile feedback thus limiting hand grasp capabilities. The proposed hand is designed to dexterity of traditional prosthetic hands while maintain the dimension and weight. Our approach is aimed at providing enhance grabby abilities and sensor motor sensor motor coordination to the amputee, by integrating miniature mechanisms, sensor, embedded control.

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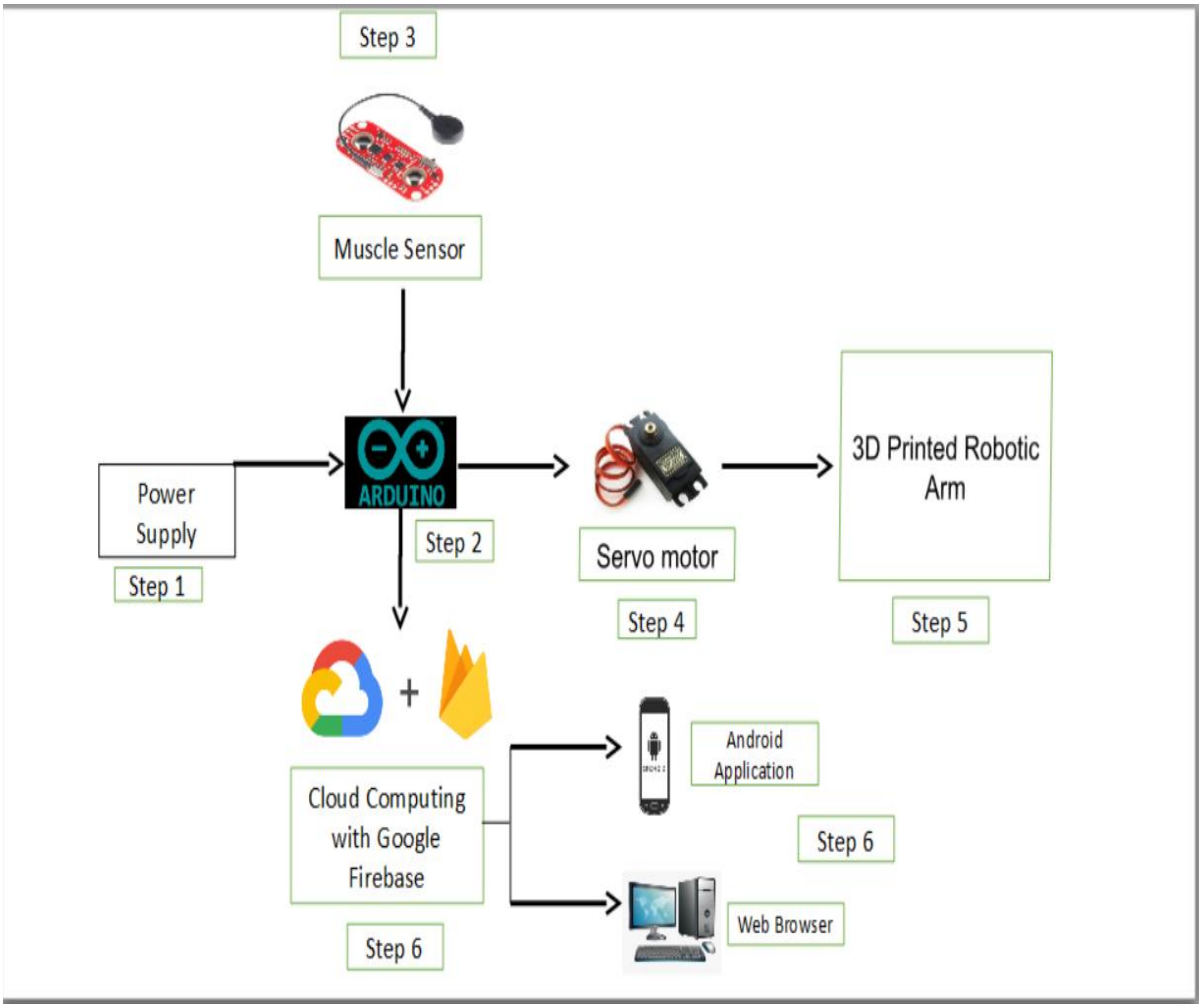
INTRODUCTION

The human hand is powerful tool for sensing and operating in the environment. In everyday life we see some people who lost their hand in accident or were born without a hand. Such people always feel lack of their non-physical organs, so the “Prosthesis Body Arm” project has been made for people who do not had arms. The prosthesis body arm is an artificial robotic hand that is not just normal prosthetic limb for people with disorder but is a robotic prothesis arm that can be used in daily life, so that those people do not feel the lack of their non-arm. This means the robotic hand should be felt by user as part of his/her own body and it provide same function of natural hand

METHODOLOGY

Arduino is a brain of a system. It works or gives input/output instructions to respected hardware and software. EMG stands for Myography, it counts muscle power of human. Cloud computing with google firebase is the delivery of computing services including server, storage, database, networking, software, analytic sand intelligence over the internet



Block Diagram

- **Step 1(Power Supply):** It provides Supply to your system.

- **Step 2(Arduino):** It is a brain of a system. It works or gives input/output instructions to respected hardware and software

- **Step 3(Muscle Sensor):** According to the values of electromyography sensor, the servo motors of prosthesis will move. It counts the muscle power of patient.

- **Step 4(Servo Motor):** Servo allow the shaft to be positioned at various angles, usually between 0 and 180 degrees.

- **Step 5(Android Applications and Web Browser):** It displays the information transferred by Cloud Computing. By using Android applications and web browser nurse, doctor, relatives and second doctor consultant can monitor patient.

- **Step 6(Alert System):** In case Patient Health is increased or decreased the blub glows according to the requirement of Alert System.

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ADVANTAGES

- Prosthesis body arm gives happiness to the person with various disabilities. They get motivation to leave their life.
- It provides movements opening and closing fingers.
- It is flexible in functioning.
- It is very easy to learn and use.

LIMITATION

- Materials used lead to skin irritation, inflammation, infection
- Frequent use of this prosthesis device requires replacement of the entire arm in few years (about 3-5 years)

FUTURE SCOPE

- Artificial arm is essential to improve the quality of life of people living without hands.
- This technology still causes adverse events

CONCLUSION

Our prosthesis body arm is useful for handless people. Which is used for their daily life and it gives happiness and get motivation to leave their life.

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REFERENCES

1. https://www.researchgate.net/publication/51759015_Bionic_prosthetic_hands_A_review_of_present_technology_and_future_aspirations.